Customer Churn Prediction Report

Executive Summary

The results of the churn prediction research utilizing two models, Logistic Regression (Model 1) and Random Forest (Model 2) are presented in this report. According to the compared findings, Random Forest continuously performed better than Logistic Regression in every important assessment metric, such as accuracy, precision, recall, F1-Score and ROC AUC. Random Forest is the suggested model for deployment due to its exceptional performance.

The study also examines how the chosen model should be modified to operate with the SS1 dataset, provides suggestions for enhancing it and assesses how well the results agree with the body of literature covered in Part A. The suggestions focus on useful enhancements such as managing class imbalance, feature engineering and hyperparameter tweaking. The significance of retraining with SS1-specific features and preserving repeatability is emphasized by adaptation techniques. Lastly, the results are demonstrated to be consistent with other studies that consistently indicate how successful ensemble approaches are in churn prediction.

1. ***Best Model***

   The comparison of Random Forest and Logistic Regression shows that Random Forest is the more powerful model. Accuracy of 0.818, precision of 0.612, recall of 0.201, F1-Score of 0.303 and ROC AUC of 0.585 were all attained with logistic regression. Random Forest, on the other hand, obtained an accuracy of 0.861, precision of 0.784, recall of 0.404, F1-Score of 0.534 and ROC AUC of 0.689.

   The most significant difference lies in recall. In a business context, recall is crucial because missing churn cases results in unintervened client loss. Proactive retention techniques are made possible by Random Forest's higher recall, which guaranties that more at-risk clients are identified.

   Random Forest offers feature relevance insights in addition to metrics, emphasizing factors like Age and Number of Products as important predictors. The client benefits from this interpretability since it helps with focused marketing and retention initiatives.

Therefore, the suggested recommendation would be the adoption of Random Forest as the primary churn prediction model, as it offers superior predictive power and actionable insights.

2. ***Recommendations for Improvement***

   Although Random Forest outperforms Logistic Regression, there are opportunities to further enhance its performance.

   Hyperparameter Tuning

   The default settings were used to train the present Random Forest model. Accuracy and recall can be increased by using grid search or cross-validation to optimize hyperparameters like the number of trees, maximum depth and minimum samples per split.

   Feature Engineering

   Predictions are dominated by Age and Number of Products, according to the feature importance analysis. Although helpful, this implies that other behavioural patterns might not be adequately captured by the model. Developing new features, such tenure per product, balance-to-salary ratios or interaction terms, can offer deeper insights into consumer behaviour.

   Class Imbalance Handling

   There are fewer churn cases in the dataset than non-churn cases. Recall is impacted by this imbalance. Without compromising accuracy, methods like the Synthetic Minority Oversample Technique (SMOTE), undersampling or modifying class weights might increase sensitivity to churn cases.

   Alternative Algorithms

   Even though Random Forest works well, investigating Gradient Boosting or XGBoost might produce even better results. In churn prediction tasks, these systems frequently attain superior recall and F1-scores.

   Model Monitoring

   Consumer behaviour changes throughout time. To maintain accuracy, regular retraining and performance monitoring of the model are crucial. Maintaining relevance can be achieved by setting up a retraining plan such as quarterly.

Business Insight

Enhancing recall guaranties that fewer at-risk clients are overlooked, allowing for focused retention initiatives. Improved accuracy lowers false positives, preventing resources from being squandered on clients who are unlikely to leave.

3. *Adaptation for SS1 Data*

To apply the Random Forest model to the SS1 data, several adaptions are required.

Preprocessing Adjustments

The data request from SS1 may contain new categorical variables that need to be encoded, characteristics that need to be scaled and possibly missing values that need to be imputed. To account for these modifications, SS2's preprocessing pipeline needs to be expanded.

Integration of SS1-Specific Features

Requested elements including product usage trends and customer interaction indicators are included in the data. To guarantee that projections accurately represent SS1's distinct clientele, these factors must be included in the model.

Retraining and Validation

To guarantee that performance indicators stay high, the Random Forest model needs to be retrained on SS1 data and then validated. To verify generalizability, cross-validation should be employed.

Adapting the model ensures that churn predictions are tailored to SS1's specific customer context, enabling more accurate and actionable insights.

4. *Alignment with Literature*

The study's findings are in good agreemenr with those reported in the literature review. In customer churn prediction, ensemble techniques like Random Forest, Gradient Boosting and XGBoost regularly outperform more straightforward classifiers like Logistic Regression across a number of articles.

Imani et al. (2025) conducted a thorough analysis of recent developments in churn prediction and came to the conclusion that ensemble approaches are the

most popular, with deep learning techniques like CNNs and LSTMs showing promise as alternatives. Their focus on preprocessing methods, such as category encoding, feature scaling and balancing unbalanced datasets, is similar to the approach used in this study. Their conclusion that ensemble models produce better prediction results is supported by Random Forest's superior performance in our results (accuracy 0.861, recall 0.404).

Peng et al. (2023) emphasized the significance of interpretability and showed that ensemble approaches perform better than simple classifiers. They showed how Random Forest may offer business insights beyond simple forecasts by using SHAP analysis. This focus on interpretability and practical insights for business decision-making is reflected in our feature importance study, which found Age and Number of Products as important predictors.

Mouli et al. (2024) compared several classifiers and discovered that Random Forest and SVM had the best performance whereas Logistic Regression was still comprehensible but less accurate. This is exactly in line with our results, which showed that Random Forest outperformed Logistic Regression in terms of recall (0.201) and F1-score (0.303).

Finally, the Badol (2024) Kaggle Banking Customer Churn Dataset confirms the significance of the dataset we selected. To ensure methodological consistency with existing studies, our approach likewise used the preprocessing procedures described in the literature, including feature selection, encoding, normalization and class imbalance handling.

References

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